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Sta 301

imp Questions

1 A sample of size $n=3$ drawn without replacements experiment from a population $N=5$ item whose values 0, 2, 3, 6, 7 draw possible samples.

Sol:- Sample without replacement $= \binom{N}{n}$

$= \binom{5}{3} = 10$

(0,2,3) (0,2,6) (0,2,7) (0,3,6) (0,3,7)

(0,6,7) (2,3,6) (2,3,7) (3,6,7)

2 Calculates sampling error if Sampling / sample mean is 102 & population mean is 100.

Sol:- Sample mean $= \bar{X} = 102$

population mean $= \mu = 100$

Sampling error $= \bar{X} - \mu$

$= 102 - 100 = 2$

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lect 23 to 45.

3 Let x be a random variable with probability distribution.

x	f(x)/p(x)	x.p(x)
-1	0.125	-0.125
0	0.50	0
1	0.20	0.20
2	0.05	0.10
3	0.125	0.375
	1	0.55

$\mu = E(x) = \Sigma x \cdot p(x) = 0.55$

2 From the following joint probability dist, find g(2).

y|x	2	4	h(y)
$y_1 = 1$	0.10	0.15	0.25
$y_2 = 3$	0.20	0.30	0.50
$y_3 = 5$	0.10	0.15	0.25
g(x)	(0.40)	0.60	1.00

thus,

$x = 2, \quad g(2) = 0$

Let x be a continous random var.
with probability d. f.

$$f(x) = \begin{cases} 6x(1-x), & 0 \le x \le 1 \\ 0 & \text{otherwise.} \end{cases}$$

1. non negatively:-

$f(x) = 6x(1-x) \quad 0 \le x \le 1$

bothe are non negative.

$6x = 0, \; 1-x = 0 \Rightarrow x = 0, \; x = 1$

2. chek normalization:-

$$\int_{-\infty}^{\infty} f(x)\,dx = 1 \quad 0 \le x \le 1$$

$$\int_0^1 6x(1-x)\,dx \Rightarrow 6\int_0^1 x(1-x)\,dx$$

$$6\int_0^1 (x - x^2)\,dx \Rightarrow 6\int_0^1 x\,dx - \int_0^1 x^2\,dx$$

$$= 6\left[\left|\frac{x^2}{2}\right|_0^1 - \left|\frac{x^3}{3}\right|_0^1\right]$$

$$= 6\left[\frac{(1)^2}{2} - \frac{(0)^2}{2}\right] - \left[\frac{(1)^3}{3} - \frac{(0)^3}{3}\right]$$

$$= 6\left[\frac{1}{2} - \frac{1}{3}\right] \Rightarrow 6\left[\frac{3-2}{6}\right] \Rightarrow 6\left(\frac{1}{6}\right)$$

$\Rightarrow 1$

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4 Quartile deviation

$$Q.D = \frac{Q_3 - Q_1}{2}$$

5 Find mean & variance for the sampling distribution given below.

$\hat{p}$	no. of samples	$f(\hat{p})$	$\hat{p} \cdot f(\hat{p})$	$\hat{p}^2 \cdot f(\hat{p})$
0	1	1/20	0	0
1/3	9	9/20	3/20	1/20
2/3	9	9/20	3/20	1/5
1	1	1/20	1/20	1/20
Total	20	1	1/2	3/10

$$\mu_{\hat{p}} = \sum \hat{p} \cdot f(\hat{p}) = \frac{1}{2} = 0.5$$

$$E(x^2) - (E(x))^2$$

$$\sigma^2(\hat{p}) = \sum \hat{p}^2 \cdot f(\hat{p}) - \left[\sum \hat{p} \cdot f(\hat{p})\right]^2 = \frac{3}{10} - \left(\frac{1}{2}\right)^2$$

$$\sigma^2(\hat{p}) = \frac{3}{10} - \frac{1}{4} \Rightarrow \frac{1}{20} = 0.05$$

$$\sqrt{\sigma^2} = \sqrt{0.05} \Rightarrow \sigma$$

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6 If $n = 10$ & variance of the binomial distribution is 21.25 then find the standard deviation of the binomial distribution.

Sol:- ~~Sol~~ Formula:-

Binomial distribution $= npq$ $\quad q = 1-p$

$S.D = \sqrt{npq} = npq$ $\quad \because p =$ success

$\because n =$ no. of trials.

Variance $= 21.25$

Number of trials $= n = 10$.

Variance $= 10pq$ $\quad (n\ p\ q)$

$10p(1-p)$.

$21.25 = 10p - 10p^2$

$10p^2 - 10p + 21.25 = 0$

$\because ap^2 + bp + c = 0$

$a = 10$, $b = -10$, $c = 21.25$

$ax^2 + bx + c = 0$

$$p = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$P = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(10)(21.25)}}{2(10)}$$

$$P = \frac{10 \pm \sqrt{100 - 850}}{20}$$

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$$p = \frac{10 \pm \sqrt{-750}}{20}$$

$\Rightarrow \sqrt{-750}$ Since a binomial distribution should not result in a negative value under square roots, mesning there may be a mistake with the provided variance.

=> Standard deviation of binomial distribution:-

$$\sigma = \sqrt{\text{variance}} = \sqrt{21.25} \approx 4.61$$

7. A computer shop contains 3 good & 2 defective monitors. two monitors are select by without replacement Find the probability. that.

i) Both will be good.

ii) One is good & one is defective.

Sol:- Given:-

3 good monitors

2 defective monitors

Total monitors is 5.

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Sample space = ${}^{n}C_r = {}^{5}C_2 = 10$

∵ combination formula = ${}^{n}C_r = \binom{n}{r}$

$\binom{5}{2} = \binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{5!}{2!(5-2)!}$

$\frac{5.4.3.2.1}{2! \times 3!}$

i) Both are good.

$= \frac{{}^{3}C_2 \times {}^{2}C_0}{{}^{5}C_2} = \frac{3 \times 1}{10} = \frac{3}{10}$

ii) One is good & one is defective:-

$= \frac{{}^{3}C_1 \times {}^{2}C_1}{{}^{5}C_2} = \frac{3 \times 2}{10} = \frac{6}{10}$

8 If $P(x=x) = \binom{6}{x}\left(\frac{1}{2}\right)^{x}\left(\frac{1}{2}\right)^{6-x}$.

then find $P(x \geq 5)$.

$P(x \geq 5) =$

$\Rightarrow \binom{6}{5}\left(\frac{1}{2}\right)^{5}\left(\frac{1}{2}\right)^{6-5}$

$= \binom{6}{5}\left(\frac{1}{32}\right)\left(\frac{1}{2}\right)^{1} = \left(\frac{6}{320}\right)$

$= 6 + 320 = 326$ A.

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Ref 31 to 35.

9 Calculate mean & standard deviation from the given below sampling distribution of sample mean.

Sol:-

$\bar{x}$	$f(x)$	$x \cdot f(x)$	$x^2 \cdot f(x)$
14/3	1/4	7/6	49/9
16/3	2/4	8/3	128/9
17/3	1/4	17/12	289/36
	4/4=1	5.25	27.69

Sampling distribution of sample means-

$$\mu_{\bar{x}} = \Sigma(x \cdot f(x)) = 5.25$$

$$\text{Variance} = \sigma^2 = \Sigma x^2 f(x) - (\Sigma x \cdot f(x))^2$$

$$= 27.69 - (5.25)^2$$

$$= 27.69 - 27.5625$$

$$\sigma^2 = 0.1275.$$

Square root on b/s for S.D.

$$\sqrt{\sigma^2_{\bar{x}}} = \sqrt{0.1275}$$

$$\sigma = 0.36 \text{ Ans.}$$

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10 Two dice are thrown. Make the Sample space & also find the number of sample points in an event A that the sum is 7.

Sol:

$$S = \begin{Bmatrix} (1,1)(1,2)(1,3)(1,4)(1,5)(1,6) \\ (2,1)(2,2)(2,3)(2,4)(2,5)(2,6) \\ (3,1)(3,2)(3,3)(3,4)(3,5)(3,6) \\ (4,1)(4,2)(4,3)(4,4)(4,5)(4,6) \\ (5,1)(5,2)(5,3)(5,4)(5,5)(5,6) \\ (6,1)(6,2)(6,3)(6,4)(6,5)(6,6) \end{Bmatrix}$$

$$A = \{(1,6)(6,1)(2,5)(5,2)(4,3)(3,4)\}$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{6}{36} = \frac{1}{6} \text{ Ans.}$$

11 Write down the formula of combined or pooled proportion of two samples.

$$P_c = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

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12 If approximate value of class interval is 2.96 & range = 14.8 then find the number of classes.

Sol:- class interval = $\frac{\text{range}}{\text{number of classes.}}$

$$2.96 = \frac{14.8}{\text{no. of classes.}}$$

$$\text{no. of class} = \frac{14.8}{2.96} = 5.$$

13 Find the coefficient of variation (C.V) for the following price of commodity

Sol:-

x	x^2
8	64
13	169
18	324
23	529
30	900
$\Sigma x = 92$	1986

$$\bar{x} = \frac{\Sigma x}{n} = \frac{92}{5} = 18.4$$

$$S = \sqrt{\frac{\Sigma x^2}{n} - \left(\frac{\Sigma x}{n}\right)^2}$$

$$S = \sqrt{\frac{1986}{5} - \left(\frac{92}{5}\right)^2}$$

$$S = \sqrt{58.64} \Rightarrow 7.66$$

$$C.V = \frac{S.D}{\bar{x}} \times 100$$

$$C.V = \frac{7.66}{18.4} \times 100 = 41.63\%$$

14 Probability of one electric device failure is 0.01. If a sample is chosen of 400, what is probability that exactly 2 are defective.

Sol: $p = 0.01$, $q = 1-p \Rightarrow 1-0.01 \Rightarrow 0.99$

$$P(X=2) = \binom{400}{2} (0.01)^2 (0.99)^{400-2}$$

$$= 79800 \times 0.0001 \times 0.02$$

$$= 0.1596$$

15 If X is binomial distribution $n=5$ $p=0.5$, $q=0.5$ then find SD(X).

Sol: $S.D(X) = \sqrt{npq} =$

$$S.D(x) = \sqrt{5 \times 0.5 \times 0.5} = \sqrt{1.25}$$

$$= 1.12$$

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16 Two dice are rolled. Find probability that outcome is equal or more than 11.

Sol $S = \{(1,1)(1,2)(1,3)(1,4)(1,5)(1,6)$
$(2,1)(2,2)(2,3)(2,4)(2,5)(2,6)$
$(3,1)(3,2)(3,3)(3,4)(3,5)(3,6)$
$(4,1)(4,2)(4,3)(4,4)(4,5)(4,6)$
$(5,1)(5,2)(5,3)(5,4)(5,5)(5,6)$
$(6,1)(6,2)(6,3)(6,4)(6,5)(6,6)\}$

$A = \{(6,5)(5,6)(6,6)\}$

$P(A) = \frac{n(A)}{n(S)} = \frac{3}{36} = \frac{1}{12} = 0.83.$

17 $E(xy) = 421$, $E(x) = 42$, $E(Y) = 15$
check independence.

$Cov(X,Y) = E(XY) - E(X)E(Y)$
$= 421 - (42)(15)$
$= -209$
It is not independent.

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18 If $E(X) = 0.5$ then find $E(10.5+2Y)$

Sol:- $E(10.5+2Y)$

$= 10.5 + 2E(Y)$

$= 10.5 + 2(0.5)$

$= 11.5$ ★.

19 If $E(x) = 0.7$ then find $E(2x)$?

Sol:- $E(2x)$

$= 2E(x)$

$= 2 \times 0.7 = 1.4$ ★.

20 Bayes theorem:-

Formula:-

$$P(A_i/B) = \frac{P(A_i)\,P(B/A_i)}{\sum_{i=1}^{k} P(A_i)\,P(B/A_i)}$$

21 If $E(XY) = 0.45$, $E(X) = (0.50)$, $E(Y) = 0.90$ then x & y are

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independent?

Sol:- $Cov(X.Y) = E(XY) - (E)(X)E(Y)$

$= 0.45 - (0.50)(0.90)$

$= 0.45 - 0.45$

$= 0$ it is independent.

22 If mean of a distribution is 0.925 & S.D = 0.132 Find out the limits by applying chebychev's inequality for k = 2. How much fraction of data will lie b/w the two limits?

$P(\mu - \sigma k \leq x \leq \mu + \sigma k) \geq 1 - \frac{1}{k^2}$

Sol:- $(\bar{X} - Sk, \bar{X} + kS)$

$= (0.925 - 2(0.132), 0.925 + 2(0.132))$

$= (0.925 - 0.264, 0.925 + 0.264)$

$= (0.661, 1.189)$

95% of the measurement will fall within these limits.

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23 What is the probability that a poker hand of 5 cards contain exactly 2 aces (hyper geometric distribution)?

$$P(X=2) = \frac{\binom{4}{2}\binom{48}{3}}{\binom{52}{5}}$$

$$= \frac{6 \times 17296}{2598960} = \frac{103776}{2598960}$$

$$= 0.0399$$

24 Hyper geometric distribution formula:-

$$P(X=x) = \frac{\binom{k}{n}\binom{N-k}{n-x}}{\binom{N}{n}}$$

N = number of units in population

n = number of units in sample

k = number of success

parameters. (N, n, k)

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25 If $x = 327$, $n = 634$, $p_0 = 0.50$ then find the z test statistic for proportion.

$$z = \frac{x \pm 1/2 - np_0}{\sqrt{np_0(1-p_0)}}$$

$\Rightarrow np_0 = 634 \times 0.50 = 317$

here, $x > np_0$ therefore we use $x - \frac{1}{2}$

$$z = \frac{x - 1/2 - np_0}{\sqrt{np_0(1-p_0)}}$$

$$z = \frac{327 - 1/2 - 317}{\sqrt{317(1-0.50)}}$$

$$z = \frac{9.5}{12.58} = 0.75$$

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26 If population proportion are given as

$P_1 = 0.30$, $P_2 = 0.20$

$\delta^2 P_1 - P_2$ where $n = 10$

Sol:- $\sigma_{\hat{P}_1 - \hat{P}_2} = \frac{P_1 q_1}{n_1} + \frac{P_2 q_2}{n_2}$

$q_1 \Rightarrow 1 - P_1 \Rightarrow 1 - 0.30 = 0.7$

$q_2 \Rightarrow 1 - P_2 \Rightarrow 1 - 0.20 = 0.8$

$\sigma_{\hat{P}_1 - \hat{P}_2} = \frac{(0.3)(0.7)}{10} + \frac{(0.20)(0.8)}{10}$

$= 0.021 + 0.016$

$= 0.037$ Ans.

27 Binomial distributions-

$P(X = x) = \binom{n}{x} p^x q^{n-x}$

Parameter = n, p

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28 Define an unbiased estimater.

An unbiased estimater is a statistical estimater that, the expected value of the unbiased estimator is equal to the parameter being estimated.

Formula:-

$$E(\hat{\theta}) = \theta$$

$\hat{\theta}$ is an estimator of a parameter θ

$\hat{\theta}$ = unbiased.

Where $E(\hat{\theta})$ denotes the expected value of $\hat{\theta}$.

29 What is meant by confidence interval.

A confidence interval is a range of values drived from sample data, that is ----------

Formula:- $\bar{x} \pm z\left(\frac{\sigma}{\sqrt{n}}\right)$

∴ Population mean = μ

∴ Sample mean = $\bar{x}$

∴ Standard deviation = σ

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∴ Sample size = n

Where z is the critical value from the standard ~~deviation~~ normal distribution corresponding to the desired confidence level.

3. Define one tailed & two tailed tests?

Sol:- two tailed tests:-

- Null hypothesis μ = given
- Alternate hypothesis = $\mu \neq 0$

⇒ One tailed tests:-

- Null hypothesis = μ = given.
- Alternate hypothesis = $\mu > 0, \mu < 0,$ given.